

SOUTH CENTRAL RLY

**HYDERABAD
DIVISION-S AND T
2nd Floor, S and T
Department,
Hyderabad Bhavan,
Secunderabad.
Secunderabad,
500017
Telangana, India**

**Letter No: HYDERABAD DIVISION-S AND T / Y-SG-
36-2024-25-12 / 01052610118677**

Dated: 05/12/2024

**M/s NAGALAKSHMI ENTERPRISES-
SECUNDERABAD**

PLOT No 172 RAVI COLONY MAHENDRA HILLS
EAST MARREDPALLY SECUNDERABAD
SECUNDERABAD- 500026
Telangana, India

Sub: Letter Of Acceptance

- Ref:**
1. Tender No. Y-SG-36-2024-25-12 closing date 14-11-2024 11:30 for Hyderabad Division-Carrying out Repairs to Telecom Assets on Secunderabad (Excl)-DHNE (Excl) and Secunderabad (Excl)-MUE (Excl) section and at Hqrs area, Moula-ali, Lallaguda & Secunderabad areas
 2. Your bid ID **17544643** dated **14/11/2024 10:16**

The Competent Authority has accepted your offered rates in connection with the subject work. The total cost of the work at the accepted rates works out to Rs. 20485085.79 (Rupees Two Crore Four Lakh Eighty-Five Thousand And Eighty-Five Rupees And Seventy-Nine Paise Only)

A sum of Rs.257400 deposited as Earnest Money vide IREPS reference ID PE927224356709 has been retained towards initial Security Money for due and faithful fulfillment of the contract, and the balance Security Money will be recovered from the progressive bills @ 6 % of the bill amount till it reaches 5 % of the contract value in terms of clause 16(1) of GCC.

You are requested to submit Performance Guarantee in the form as given in Clause 16(4) of GCC equivalent to 5 % of the contract value amounting to Rs. 1024254.29 (Rupees Ten Lakh Twenty-Four Thousand Two Hundred And Fifty-Four Rupees And Twenty-Nine Paise Only) within 21 days from the date of issue of Letter of Acceptance, valid up to stipulated date of completion plus 60 days so that contract agreement can be executed.

The entire work shall be completed within 12 month from the date of issue of Letter of Acceptance.

Performance Guarantee (PG): The successful bidder shall have to submit a Performance Guarantee (PG) within 21 (Twenty one) days from the date of issue of Letter of Acceptance (LOA). Extension of time for submission of PG beyond 21 (Twenty one) days and upto 60 days from the date of issue of

LOA may be given by the Authority who is competent to sign the contract agreement. However, a penal interest of 12% per annum shall be charged for the delay beyond 21(Twenty one) days, i.e. from 22nd day after the date of issue of LOA. Further, if the 60th day happens to be a declared holiday in the concerned office of the Railway, submission of PG can be accepted on the next working day.

a): In all other cases, if the Contractor fails to submit the requisite PG even after 60 days from the date of issue of LOA, the contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit Earnest Money Deposit and other dues payable against that contract. In case a tenderer has not submitted Earnest Money Deposit on the strength of their registration as a Startup recognized by Department of Industrial Policy and Promotion (DIPP) under Ministry of Commerce and Industry, DIPP shall be informed to this effect. The failed Contractor shall be debarred from participating in re-tender for that work. The format for submission of Bank Guarantee is enclosed. This PG shall be initially valid up to the stipulated date of completion plus 60 days beyond that.

b): Performance Guarantee to be deposited in favor of SENIOR DIVISIONAL FINANCE MANAGER/ HYDERABAD DIVISION, SOUTH CENTRAL RAILWAY, SECUNDERABAD.

1) Section of work: HQ Area, SC-MUE & SC-DHNE Sections

2) Officer in charge: DSTE/P&P/HYB, ADSTE/HYB, ADSTE/South/HYB & ADSTE/NZB

3) Supervisor in charge: SSE/Tele/Division, SSE/Tele/GM Compound, SSE/Tele/DRM Compound, SSE/Tele/RN, SSE/Tele/LGD, SSE/Tele/KCG, SSE/Tele/MBNR, SSE/Tele/GWD, SSE/Tele/KRNT, SSE/Tele/MJF, SSE/Tele/MED & SSE/Tele/NZB

4) Stores-receipts/issues: SSE/Tele/Stores/LGD

5) Materials inspected by Receipts/issues: Offi: ADSTE/HYB

6) Materials inspected by Receipts/issues: Supe: Supervisor :SSE/Tele/Stores/LGD

7) Recording of measurements for Supply Portion: Officer : ADSTE/HYB

8) Recording of measurements for Supply Portion: Supervisor : SSE/Tele/Stores/LGD

9) Recording of measurements for Labour Portion: Officer : DSTE/P&P/HYB, ADSTE/HYB, ADSTE/South/HYB & ADSTE/NZB

10) Recording of measurements for Labour Portio: Supervisor : SSE/Tele/Division, SSE/Tele/GM Compound, SSE/Tele/DRM Compound, SSE/Tele/RN, SSE/Tele/LGD, SSE/Tele/KCG, SSE/Tele/MBNR, SSE/Tele/GWD, SSE/Tele/KRNT, SSE/Tele/MJF, SSE/Tele/MED & SSE/Tele/NZB

11) Bill paying authority: Sr.DFM/HYB

All Other terms and conditions, as stipulated in the tender documents shall be applicable.

FIRDAUS KHAN
DSTE/P P/HYB
Digitally Signed
[View Signature Details](#)

Awarded Quantities And Rates

Item Sno.	Item Desc	Item Code	Item Qty	Qty Unit	Unit Rate(Rs)	Escl. (%)	Advt.Value (Rs)	Bid Rate/ Unit Rate	Bid Amount (Rs)
Schedule A-Schedule - A (Item Directory - Not Applicable)							2754652.00		
1	SUPPLY PORTION (SOR Items)		View Details		At Par		2754652.00	4.59 % Below	2628213.47
Schedule Totals							2628213.47		
Schedule B-Schedule - B (Item Directory - Not Applicable)							3497920.56		
1	LABOUR PORTION - (SOR Items)		View Details		At Par		3497920.56	4.59 % Below	3337366.01
Schedule Totals							3337366.01		
Schedule C-Schedule - C (Item Directory - Not Applicable)							6150023.00		
1	SUPPLY PORTION (Non-SOR Items).		View Details		At Par		6150023.00	4.59 % Below	5867736.94
Schedule Totals							5867736.94		
Schedule D-Schedule -D (Item Directory - Not Applicable)							9067990.12		
1	Labour PORTION (Non-SOR items)		View Details		At Par		9067990.12	4.59 % Below	8651769.37
Schedule Totals							8651769.37		
Total Value							21470585.68		20485085.79
							Rebate on Total Value (%)		0.00
							Net Bid Value		20485085.79

Item Breakup

Schedule	A-Schedule - A					
Item - 1	SUPPLY PORTION (SOR Items)					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply of HDPE telecom duct with outer diameter of 40mm and inner diameter of 33mm permanently lubricated as per Specn. No. RDSO/SPN/TC/45/2006 Revision 1.0 or latest or as per specification TEC GR No.G/CDS-08/02,NOV 2004 or latest . HDPE pipe shall be supplied along with 4mm dia polypropylene rope of approved quality inside wherever required. The length of HDPE telecom duct should be 1 KM. End plugs to be provided along with drum (free of cost) (2 No. per Km.).	Kilometre	13	45226.00	587938.00
2	2	Supply of HDPE duct associated accessories and tools as per the specification of NS.1: (i) Plastic couplers - Slip fit / push type for HDPE duct (2 No. per Km.) ii) End plug for HDPE duct (2 No. per KM).	Set	130	110.00	14300.00

		iii) Cable sealing plugs for HDPE duct (1 No. of per Km.)				
3	3	Supply of 50 pair cable termination MS BOX/ Block for terminating 50 pair cable containing 5 Nos. of 10 pair LSA Plus termination strips with rubber gronet with universal keys. Make: Krone/Bestnet or better	Numbers	20	1733.00	34660.00
4	4	Supply of 20 pair cable termination MS BOX/ Block for terminating 20 pair cable containing 2 Nos. of 10 pair LSA Plus termination strips with rubber gronet with universal keys. Make: Krone/Bestnet. or better	Numbers	20	1047.00	20940.00
5	5	Supply of 100 pair cable termination MS BOX/Block for terminating 100 pair cable containing 10 Nos. of 10 pair LSA Plus termination strips with rubber gronet with universal keys. Make: Krone/Bestnet. or better	Numbers	12	2625.00	31500.00
6	6	Supply and installation of 45mmx 45mm PVC channels for drawing switchboard cables at OFC/UTP cable and at stations as indicated by Railway Engineer at site.(All materials for fixing/clamping to the wall for indoor wiring purposes and drawing of indoor cables to be arranged by contractor) (Ganapati/Shreeram/star or similar make)	Metre	600	66.00	39600.00
7	7	Supply of one quad V.F. Transformer assembly (2T) to RDSO Spec. No. IRS: TC. 76 -2000 with Amdt.1 (or latest) with impedance ratio 470:1120 Ohms. (To be procured from RDSO approved firm)	Numbers	140	707.00	98980.00
8	8	Supply of VF Tapping Transformer 600:1120 Ohms to RDSO Spec No. IRS.TC.22-76 or latest (To be procured from RDSO approved firm)	Numbers	140	609.00	85260.00
9	9	Supply of Thermo shrinkable jointing kit as per Spec. No. IRS TC. 77/2012 (Rev 3.0) Amendment.1 to 3 or latest for all types joints suitable for 6 Quad Jelly filled UG cable.(To be procured from RDSO approved firms).	Numbers	220	1923.00	423060.00
10	10	Supply of desk type 2 wire , 12 way DTMF selective calling group telephone complete as per specification IRS. TC. 80-2000, Amendment-1. or latest (to be produced from RDSO approved form)	Set	20	4257.00	85140.00
11	11	Supply of Four Wire DTMF way station equipment' to RDSO spec. No. TC 60 2007 Amdt-1 or latest with telephone to RDSO spec. No.TC 38 97 or latest. to be produced from RDSO approved form	Numbers	20	3833.00	76660.00
12	12	Supply of basic material to construct unit maintenance free earth consisting as per RDSO spec no.RDSO/SPN/197 ver.1.0 or latest consisting of (To be Procured from RDSO approved firm) as per bonding connections drawing no.SDO/RDSO/E&B/002. a) copper bonded steel electrode of 3.0m (min) long, 17.0 mm dia (min) with copper bonding thickness mini 250 micron and UL listed and marked--1 no. b) Earth enhancement compound of RDSO approved brand supplied in sealed bags of min 30-35 Kgs c) Copper strip of 200X25x6mm to exothermically welded to earth electrode-1 No. d) Copper strip of 300X25x6mm (MEEB)-1 No. e) Copper strip of 150X25x6mm (SEEB)-1 nos f) Copper Tape of 25x2mm or 8mm dia-1 nos	Set	22	16837.00	370414.00
13	13	Supply of 35 Sq.mm Multi-strand single core PVC insulated copper cable as per IS:694 for connecting	Metre	1600	303.00	484800.00

		Main earth electrode to MEEB in the equipment room in duplicate.				
14	14	Supply of 10 Sq mm Multi strand single core PVC insulated copper cable as per IS:694 for connecting various equipments to SEEB.	Metre	1000	87.00	87000.00
15	15	Supply of 16 Sq mm Multi-stand single core PVC Insulated copper cable as per IS:694 for connecting MEEB to SEEB and SPDs and SPDs to MEEB.	Metre	600	144.00	86400.00
16	16	Supply of Two conductor disconnect terminal blocks (wago terminals) for test and measurement, suitable for various telecom cable terminations as per the site requirements	Numbers	6000	38.00	228000.00
					Total	2754652.00
Schedule	B-Schedule - B					
Item - 1	LABOUR PORTION -(SOR Items)					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Excavation of trench to a depth of 1200mm and width of 300mm at bottom in all types of soil(Normal soil/soft soil/sandy soil except in hard morum / rocky soils /bridges/ level crossings/ track crossings/ coil pits,) as per RDSO Sketch no SDO/CABLE LAYING /003 including marking of cable alignment, clearing of jungle, bushes, trial pits if required, refilling with excavated soil, ramming and consolidation after laying of cables complete to the finished item of work as directed by supervisor in-charge.	Route Kilo Meter (RKM)	12.5	51912.00	648900.00
2	2	Excavation of trench to a depth of 1200mm and width of 300mm at bottom in hard morum soil as per RDSO Sketch no SDO/CABLE LAYING /003 including marking of cable alignment, clearing of jungle, bushes, trial pits if required, refilling with excavated soil, ramming and consolidation after laying of cables complete to the finished item of work as directed by supervisor in-charge.	Route Kilo Meter (RKM)	10	67092.00	670920.00
3	3	Cutting of rock to a depth of 200 mm and width of 200 mm where rock is visible from surface to a depth more than 300 mm up to 550mm, excavation of trench with width 300mm above rock concreting 200mmx200mm after laying of cable with 1 cement: 2 sand : 4 graded stone aggregate 20mm nominal size and curing of concrete to the required number of days, refilling with excavated soil ramming and consolidation except in bridges, culverts, level crossings, track crossings, cable pits, rocky soil including marking of cable alignment, clearing of debris etc., complete to the finished item of work as directed by site in-charge. Drg No.SK/CN/NEW/3. All the material is to be arranged by contractor	RMT	800	241.00	192800.00
4	4	Trenching under Rails for crossing the track(Track crossing) and refilling of trenches as per RDSO drawing No.SDO/CABLE LAYING /009. The track crossing shall be done at right angles.The depth of cable trench shall be 1.2 Mtr. below the rail flange. The depth between top cable and the level of rail flange of the rail shall never be less than 1.0 Mtr.. The ballast shall be Stored separately while making the trench and the earth soil shall be stored separately. while re-filling the trench, the earth soil	RMT	600	141.00	84600.00

		has to be filled first to the ground level and the ballast has to be properly put in the track. this work has to be done under supervision of supervisor of S&T and supervisor of P.Way				
5	5	Track crossing by horizontal boring method based on the drawing no.DRM/J/6735 dated 7-8-03 (enclosed) and as advised by the engineer at site. This includes the supply and insertion(pushing) of GI pipe (80mm) confirming to IS.1239 heavy class in the bore drilled with the coupler between two pipes. All the joints of casing pipes should be tight and leak proof.	Metre	100	1403.00	140300.00
6	6	Breaking of metal/ tar road and refilling after laying the cables as per drawing no.SDO/CABLE LAYING/010. The surface of the road shall be concreted to match with the old surface as per instructions of site incharge.	RMT	900	142.00	127800.00
7	7	Laying different type cables in the trench. Termination covered elsewhere as per instructions of site incharge.	RMT	65000	5.00	325000.00
8	8	Jungle cutting along the cable route/location Boxes/EC post as per instructions of site incharge	RMT	20000	4.00	80000.00
9	9	Laying of HDPE duct 40/33 mm dia in the trenches and in protective works already provided.	Kilometre	18	4956.00	89208.00
10	10	Strengthening of foundations of signals, Full/Half location boxes, by carrying out the earth work around the foundation, ramming of the earth, caring out masonry work from bottom of earth work using country stones and cement masonry to prevent the earth from slipping down the bank.(All required materials are to be arranged by the contractor)	Numbers	35	2813.00	98455.00
11	11	Releasing of location boxes Full/Half/sheet iron without causing damage, stacking neatly at a centralised place near the site of work as shown by the in charge, and handing over the same to the Stores depot as per the inventory made.	Numbers	10	148.00	1480.00
12	12	Casting of foundation with 1:2:4, 20mm metal and river sand to be used, supply & erection of lattice type tripod towers of GI 15 meters of height as per instructions of site incharge. Provision of earth electrode near the tower and connect the same to tower which is to be erected above the station building/at the site shown by the site Engineer of the Railways.. Cement shall be supplied by Rlys. work includes installation of the antenna, drawing the cable and installation of the battery and 25W set and commissioning the system. 25W set, cable, antenna, battery will be supplied by Rlys.	Numbers	8	28594.00	228752.00
13	13	Transportation of materials from any Stores to Site and vice-versa including loading and unloading. - Lead up to 50 Kms (PTPK)	MT-Km	13000	7.31	95030.00
14	14	Transportation of materials from any Stores to Site and vice-versa including loading and unloading. - Lead above 50 Kms	MT-Km	35000	4.80	168000.00
15	15	Provision and laying in position cement concrete of 1 cement:3 sand:6 graded stone aggregate of 20mm normal size excluding cost of cement, centering and shuttering (including transportation of all the material) .	cum	36	2085.21	75067.56

16	16	Cement 53 grade (@ the Rate (in Rupees) of 2.2 quintals per Cum of concrete including transportation	Quintal	79.2	460.00	36432.00
17	17	Installation of 100/50/20 pair Krone terminal Box as supplied vide NS items on hard wood plank of size 8"x13"x1" plank and wiring. Hard wood plank and other materials like screws, bolts etc to be supplied by the contractor at his own cost.	Numbers	56	114.00	6384.00
18	18	Installation of single earth(maintenance free) including the following:- welding and fixing of copper tape (used as bus bar) welding of cadmium bronze cable to copper plate using material are covered in Sch . All the material supplied for mounting of copper tape, exothermic weld materials, tools and fixtures for welding copper tapes shall be arranged by contractor. a. Digging /Auguring/Boring of single earth pit of 100mm to 125 mm dia to the depth of approx. 2.8 meter. b. Insertion of 3.0 meter electrode and penetrating it at least .2 meter by pushing gently with sledge hammer where the natural soil is assumed. c. Filling of earth pit with earth enhancement compound (EEC) in slurry from and backfilling. d. Carrying out exothermic weld connection on single earth electrode to a copper tape of 150X25X6mm. e. Carrying out exothermic weld connection on the copper tape of 150X25X6mm on earth electrode for connecting two 35 Sq mm cables for redundancy.. f. The exothermic welding material for bonding the earth system shall be procured from RDSO approved firms, which includes graphite moulds, clamps, weld metal powders. g. Digging a trench to a depth of at least 0 .5 Mtr. and run the two 35 Sq mm earth conductor (procured against supply item) from earth pit to MEEB. h. Routing and termination of two 35 sq mm earth conductors in trench from earth pit to 300X25X6mm MEEB, which is to be fixed inside the equipment room with two insulation studs. i. 300X300X300 mm (inside dimension) concrete earth pit with a concrete lid, approx 50mm thick with pulling hooks, shall be provided to cover the pit. Back side of the cover will be painted in black and dare of the testing and average resistance value shall be written.	Set	22	4276.00	94072.00
19	19	Routing and connecting of all the SPDs in the equipment room to MEEB with 16 sq mm copper cable using copper lugs with stainless steel nut and bolts (for item procured against supply).	RMT	1000	23.00	23000.00
20	20	Routing and connecting all the equipments in the equipment/relay room to MEEB/SEEB with 10 sq mm copper cable using copper lugs with stainless steel nuts and bolts and fixtures. The work includes laying and routing the copper cable in PVC pipes/conduits and fixing them on wall.	RMT	1600	23.00	36800.00
21	21	Earthing and soldering of cable armour.as per instructions of site incharge (Unit : Per cable)	Numbers	800	21.00	16800.00
22	22	Blowing of armoured optic fiber cable (OFC Cable supplied by Railways) into the duct (covered in the schedule of supply portion) by using blowing machine and associated works. Before blowing the duct integrity shall be tested. If any defect is found during duct integrity test, the entire length of HDPE	Kilometre	10	7382.00	73820.00

		duct from one chamber to another chamber is to be replaced by the contractor at free of cost.				
23	23	Excavation of Jointing pit and provision of joint chamber of 1.2Meter dia. Of 60 cm height with 2-piece cover on top and bottom cover with holes for drainage (reinforcement drawing will be as per instructions of site incharge).This includes water cooling of chamber, backfilling and ramming of the trench after cable laying and consolidation of soil as well as disposal of soil as per plan as advised by site engineer.	Numbers	10	3745.00	37450.00
24	24	Straight through / derivation with/without transformer joints: Installation Thermo shrinkable jointing kits for straight through / derivation joints for LC gate, pump houses etc where VF transformers are not required and derivatin joint by using 2T trasformer (to be housed in the joint) . This includes tapping of nominated Quad at a specific location shown by the Railway Engineer at site to the jelly field cable already laid. All material including valco tapes except thermo shrinkable jointing kit shall be supplied by the contractor. (Refer Annexure for instructions on cable jointing).	Numbers	150	979.00	146850.00
					Total	3497920.56
Schedule	C-Schedule - C					
Item - 1	SUPPLY PORTION (Non-SOR Items).					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply of IPM Protection Plug provides both voltage and current protection for single pair on line side. Suitable for Krone modules . Features:•Protection is built in by 3 pole GD Tube• High operating voltage• Nominal current requirement.• Material: Polyvinyl Chloride (PVC).	Numbers	3000	65.00	195000.00
2	2	Supply of FDMS 24F, Fully loaded SC type FDMS with all accessories. Make: Raycham/commscope/MX/,Bestnet/D-link /Systimax/Legrand.	Numbers	50	7000.00	350000.00
3	3	Supply of 24F joint Enclosures to RDSO spec No RDSO/SPN/ TC/68/ 2014 Rev.1. or latest (to be procured from RDSO approved firm)	Numbers	60	3000.00	180000.00
4	4	Supply of RJ-45 Jacks and Termination with CAT-6 cable Make: D-Link.or similar	Numbers	5000	10.75	53750.00
5	5	Supply of 20 pair cable termination box. make: WAGO: containing part no:280-870(40 nos),280-374(1nos),249- 116(2nos),210-229/PW11-0000(1nos) and metallic box for out door use with locking facility(1nos) and Box(1nos).	Numbers	31	2143.00	66433.00
6	6	Supply of 10 pair cable termination box of WAGO make: containing part no:280- 870(20nos),280-374(1 nos),249- 116(2 nos),210-229/pw11-0000(1nos) and metallic box for out door use with locking facility(1nos) and Box(1nos).	Numbers	31	1439.00	44609.00
7	7	Supply of VF Transformer assembly (2T) suitable for underground telecom cable circuits 470:600 ohms as per RDSO specs:IRS/TC-76/2021 with Rev.1 or latest. (to be procured from RDSO approved firm)	Numbers	120	690.00	82800.00
8	8	Supply of 4G/5G WiFi Dongle with All SIM 4g,5g Network Support , Plug & Play Sim insert Data Card	Numbers	8	3200.00	25600.00

		as specified by site incharge.				
9	9	Supply Of GSM-4G Fixed wireless phone (FWP), VOLTE enable, called ID , Hands free , high performance detached TNC antenna , 2way -digital speaker , 8 speed dial keys, Call history ,Phone book for 500 contacts in built rechargeable battery with adapter, WIFI, Hotspot ,suitable to work with JIO 4G-volte SIM, Make Beetel 5F 4G/WIBRIDGE-4G-102/ MATRIX-SIMADO or equivalent standard make note: One sample to be got approad by consignee before bulk supply , Make :BEETEL MODEL F3-4G or better and as per instructions of site incharge	Numbers	20	5230.00	104600.00
10	10	Supply of Multi Cord Binders SIZE 16 mm Make:MX or better (1.5 mts length) as advised by supervisor in-charge.	Numbers	140	545.00	76300.00
11	11	Supply of Multi Cord Binders SIZE 22 mm Make:MX or better (1.5 mts length) as advised by supervisor in-charge.	Numbers	140	728.00	101920.00
12	12	Supply of CAT 6 UTP Cable suitable for 10BASE-T/100 BASE-TX/and 1000BASE-T (Gigabit Ethernet) .Cable should be 23AWG or better. Make: Amp/Legrand/ Molex/D-link/Bestnet/Comsope	Metre	11625	20.00	232500.00
13	13	Supply of SC-PC to SC-PC patch cards (5 mtrs length) with o db coupler Make:Dlink/Mx/ Legrand/Molex.	Pair	70	975.00	68250.00
14	14	Supply of F22 Fibre glass Colinear 6.7 dB Antena Make: Diamond (HSN Code- 85291029) or better or suitable to railway standards	Numbers	16	11682.00	186912.00
15	15	Supply of 30 mtr LMR 400 Series RF cable with connectors of reputed make and suitable to railway standards	Numbers	16	5428.00	86848.00
16	16	Supply of Cat 6 sheilded twisted pair (STP) Cables, Make: Comscope/Molex/Amp/Dlink /Legrand	Metre	4000	32.00	128000.00
17	17	Supply of 8 port unmanageable switch Make: Comscope/Molex/Amp/Dlink or Better.	Numbers	40	950.00	38000.00
18	18	Supply of single line hand free caller ID phone with speaker BEETEL M71 or latest of same model with RJ11 Roset and 3 numbers of pen toruch cells (Eveready) as specified by supervisor in-charge	Numbers	200	1310.00	262000.00
19	19	Supply of 8 Port PoE Switch as per specification given in Annexure - XVIII	Numbers	12	7965.00	95580.00
20	20	Supply of 8 Port Gigabyte Switch as per specification given in Annexure - XVIII	Numbers	16	4425.00	70800.00
21	21	Supply of 4 Way DTMF Line Cords suitable to existing DTMF equipment and as per requirement at site Make: Epsilon / Bentron or better	Numbers	900	58.98	53082.00
22	22	Supply of 2 Way Line Cords RJ-11 type jack both side for Push Button Telephone length 1.5 mtrs. Make: D-Link / MX / BEETEL. or better	Numbers	1100	9.85	10835.00
23	23	Supply of 12V/65 AH VRLA Batteries Make : EXIDE/,HBL/ AMRON	Numbers	70	5130.00	359100.00
24	24	Supply of 12V/100 AH VRLA Batteries Make: EXIDE/,HBL/ AMRON	Numbers	70	8239.00	576730.00
25	25	Supply of 12V/42 AH VRLA Batteries Make: EXIDE,HBL AMRON	Numbers	70	4352.00	304640.00

26	26	Supply of 12 volts 7 Ah VRLA Battery. Make: Exide / Amaron / HBL.	Numbers	170	894.00	151980.00
27	27	Supply of Tool kit as per specification given in Annexure - XVIII	Set	8	31434.00	251472.00
28	28	Supply of DIN Rail for fixing Wago Disconnection type terminal in network rack/location Boxes.	RMT	250	75.00	18750.00
29	29	Supply of End plate for Wago connectors	RMT	700	35.00	24500.00
30	30	Supply of Separator for Wago connectors	RMT	1500	35.00	52500.00
31	31	Supply of Push Button Telephone Handset receiver Cords 4 way RJ-11 type both ends 2 mtrs. Make: D-Link / MX / BEETEL or similar.	Numbers	700	13.85	9695.00
32	32	Supply of RG-6 PVC Copper Co-axial cablebox 305 (Rmt) (Make: Deltone / Finolex/Comscope) in the PVC Casing/ PVC/ Pipes/Channels.-per Metre	RMT	1000	42.00	42000.00
33	33	Supplyg of RG-11 PVC Copper Co-axial cable box305 Rmt (Make: Deltone / Finolex/Comscope) in the PVC Casing/ PVC/ Pipes/Channels.-per Metre	RMT	500	72.00	36000.00
34	34	Supply of single line cordless Phone Panasonic Model No. KX-TG3711SX Cordless Phone (Silver colour or as specified by site incharge) or latest	Numbers	50	2500.00	125000.00
35	35	Supply of single line Boss secretary Telephone with all accessories Make Beetel M-78 or better	Set	15	3500.00	52500.00
36	36	Supply of CAT-6 cable patch cards-2mtrs length with RJ-45 Jacks on both ends Make: D-Link/Bestnet	Numbers	100	180.00	18000.00
37	37	Supply of CAT-6 cable patch cards-3mtrs length with RJ-45 Jacks on both ends Make: D-Link/ Bestnet	Numbers	50	370.00	18500.00
38	38	Supply of CAT-6 cable patch cards-5mtrs length with RJ-45 Jacks on both ends Make: D-Link/ Bestnet	Numbers	50	560.00	28000.00
39	39	Supply of 6 Pair PVC switch board cable twisted in pairs PVC sheathed copper conductor 0.5mm dia (Make: Delton/Finolex/ Surabhi)	RMT	2500	31.00	77500.00
40	40	Supply of 10 Pair PVC switch board cable twisted in pairs PVC sheathed copper conductor 0.5mm dia (Make: Delton/Finolex/ Surabhi)	RMT	3900	48.00	187200.00
41	41	Supply of 20 Pair PVC switch board cable twisted in pairs PVC sheathed copper conductor 0.5mm dia (Make: Delton/Finolex/ Surabhi)	RMT	2900	86.00	249400.00
42	42	Supply of Dual pole Wireless ADSL2+ 4 Port Modem Router with one Wan Port and RJ11 connector Make:Dlink,TPLink,Tenda	Numbers	70	1649.00	115430.00
43	43	supply of LSA plus 2/10 disconnection modules. Make Krone/ Bestnet/Airtel	Numbers	500	186.44	93220.00
44	44	supply of HDMI Cable 1.5 Mtrs length with (Male-Female/Female-Female/Male-Male) moulded type as required at site.Make: MX/ ATEN/ Bestnet	Numbers	15	785.00	11775.00
45	45	Supply of krone tools for Telephone as specified by supervisor in-charge	Numbers	10	1850.00	18500.00
46	46	Supply of remote control unit for existing curtain opening (12vDC. Operated) of reputed make as specified by supervisor in-charge	Numbers	4	2200.00	8800.00
47	47	supply of 5 mtr MPO casset OM4 24 fiber single mode trunk cable combination of SC-SC/SC-LC/LC-	Numbers	8	9500.00	76000.00

		LC as per site requirement Make:MX/D-Link/Bestnet/Deyton				
48	48	supply of 10 mtr MPO casset OM4 24 fiber single mode trunk cable combination of SC-SC/SC-LC/LC-LC as per site requirement Make:MX/D-Link/Bestnet/Deyton	Numbers	8	13800.00	110400.00
49	49	supply of 15 mtr MPO casset OM4 24 fiber single mode trunk cable combination of SC-SC/SC-LC/LC-LC as per site requirement Make:MX/D-Link/Bestnet/Deyton	Numbers	2	17500.00	35000.00
50	50	supply of 20/25 mtr MPO casset OM4 24 fiber single mode trunk cable combination of SC-SC/SC-LC/LC-LC as per site requirement Make:MX/D-Link/Bestnet/Deyton	Numbers	4	20900.00	83600.00
51	51	supply of Fiber cassette fore 4 or 8 adapters plate Panel,Rear 2 x 12MPO male to Front 6 x LC Quad ports,single Mode OM4/OM5 Make:MX/D-Link/Bestnet/Deyton	Numbers	12	8260.00	99120.00
52	52	supply of 19"Rack mount 1RU sliding 4 cutout unloaded Fiber LIU accepts Adapter plates/Cassettes	Numbers	5	2360.00	11800.00
53	53	Supply of Crimping tool Make:Fluke/spectra	Numbers	20	1200.00	24000.00
54	54	Supply of SC-LC/ LC-LC fiber patch cords as per site requirements customised from 200 mm to 2000 mm	Numbers	150	1050.00	157500.00
55	55	Supply of 4F HFTP with SC/APC Adaptor and 4 nos-Pigtail	Numbers	20	1121.00	22420.00
56	56	Supply of Rack mountable 2X16 Optical splitter with SC/APC connector Make:R&M/Systimax/panduit	Numbers	4	8850.00	35400.00
57	57	Supply of 19" Rack Mountable 2X32 Optical Splitter with SC/APC IO Connector Make: R&M / Systimax / panduit.	Numbers	2	19890.00	39780.00
58	58	Supply of 48 Core LIU, 19" Rack Mountable, with fully loaded SC/APC IO connector and Pigtails Make: R&M / Systimax / panduit	Numbers	6	17582.00	105492.00
59	59	Supply of 2/3/4 way splitters of reputed make	Numbers	30	150.00	4500.00
					Total	6150023.00
Schedule	D-Schedule -D					
Item - 1	Labour PORTION (Non-SOR items)					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply and installation of 230V AC/12V DC SMPS charger with 10 Amps suitable to 25W VHF set Make:Axiom or better	Numbers	40	2150.00	86000.00
2	2	Supply & installation of Voltage Stabilizer, capacity: 3 KVA, 1 Ph; Input Voltage 140-270V AC; Output voltage 230V (+/-) 10%, with a double pole switch of 32 Amps. Make: Hykon/Servomax or better	Numbers	4	13632.00	54528.00
3	3	Supply and installation of Class D/C type Lightning Cum surge protection safety device with Visual indication, pluggable and testable with surge tester and potential free contacts. Outputs suitable for 48V DC applications (Conforming to RDSO Specn No. RDSO/ SPN/165/2012 or latest and should be tested as per IEC 61643) it also includes supply and installation of Surge Protection devices duly wired	Set	4	5236.00	20944.00

		mounted on din rail with necessary fittings like Nuts and bolts etc. as per the site requirements and the Engineer in charge.				
4	4	Supply and installation of 4 way Spike busters with 5 mtrs power cable with individual switch And Fused for each socket 5A Make.NAX./ Goldmedal and as per instructions of site in-charge.	Numbers	58	1200.00	69600.00
5	5	Supply and installation of 1GBPS Media Converter as specified by supervisor in-charge with 2 optical port Make:Dlink/Digid/ Team Engineer or Better.	Numbers	140	5752.50	805350.00
6	6	Supply and installation of 10/100 single mode media converter with accessories including connecting cables. Make: MRO-TEK/D-Link/RAD/Team Engineer and as specified by supervisor in-charge.	Numbers	102	4000.00	408000.00
7	7	Supply and fixing of bitumen based water proofing membrane make of Dr.FIXTPIDILET to be laid for arresting roof leakages at Prefab OFC hut.	Sqm	200	350.00	70000.00
8	8	Supply & fixing of 9U rack with locking front glass door and fully vented rear steel door, compatible with standard 19" accessories, top cover with gland plate and fan, with rack cage nuts and screws, fan assembly kit, rack mount shelves, Power manager.etc this includes wiring as per the instruction of site incharge. Make: HCL/ Vero president/Dlink/Bestnet	Numbers	22	9000.00	198000.00
9	9	Supply and laying of LC to LC Patch cords (5 mts length)with 0dB adaptors as per TEC spec G/OJC-01/03 Jan 99 or latest and as specified by supervisor in-charge Make:MX/D-Link/Bestnet.	Pair	70	1275.00	89250.00
10	10	Supply and laying of SC to LC Patch cords (5 mts length)with 0dB adaptors as per TEC spec G/OJC-01/03 Jan 99 or latest as required at site as specified by supervisor in-charge Make:MX/D-Link/Bestnet.	Pair	70	1275.00	89250.00
11	11	Supply and installation of 24F SC FDMS Complete set rack mounted 24F FMS fiberpart no.fms-1u 24FS 24FP Raychem make including 24Nos 0.2db FC/PC adaptorsand 24numbers 900 micron 1.5 meter SCPC pigtails. RDSO Spec No.: RDSO/SPN/TC-037-2000 (Ver.3) or latest. Make Raychem/Bestnet/Dlink or better	Numbers	20	12078.00	241560.00
12	12	installation and splicing of bare Optical fiber in defferent types of fiber termination boxes as per instructions of site incharge	Numbers	30	1742.47	52274.10
13	13	Supply and provision of dry sand in FRP /RCC joint chanbers up to 400mm height.	Numbers	130	575.07	74759.10
14	14	Supply and fixing of 6 quad cableTermination box of WAGO make or better. Consists of partnos:280-870(24nos), 280-374(1nos), 249-116(2nos), 210-229/PW11-0000 (1nos) and Metallic box for outdoor use with locking facility 1 nos.	Numbers	30	1533.00	45990.00
15	15	Supply & fixing of 40 mm PVC Pipes (Make: Sudhakar/Nandi/SR/Panduit or better) on the walls as required by the Railways. Necessary fixing the materials like screws, washers galvanized, GI clamps, wooden pegs teak etc are to be arranged by the contractor (3 clamps per metre to be used) as specified by supervisor in-charge.	RMT	2200	129.00	283800.00

16	16	Supply & fixing of 25 mm PVC casing (Make: Sudhakar/Nandi/SR/Panduit or better) on the walls as required by Railways at site. Necessary fixing the above like screws, washers galvanized, wooden pegs teak etc are to be arranged by the contractor (5 screws per metre to be used),as specified by supervisor in-charge.	RMT	2700	41.00	110700.00
17	17	Supply &fixing of 40mm PVC casing (Make: Sudhakar/Nandi/SR/Panduit or better) on the walls as required by Railways at site. Hardware for fixing the above screws washers galvanized wooden pegs teak etc are to be arranged by the contractor (5 screws per metre to be used),as specified by supervisor in-charge.	RMT	2550	52.00	132600.00
18	18	Supply and fixing/laying of 25mm dia 2mm thick PVC pipe (Make: Sudhakar/Nandi/SR/Panduit or better) on the walls. All the required accessories such as GI clamps, Deep boxes, Jn Boxes, Normal bends, diversion bends, Joints etc are to be arranged by the contractor,as specified by supervisor in-charge.	RMT	3500	33.50	117250.00
19	19	Supply and fixing/laying of 25mm dia 2mm thick PVC pipe (Make: Sudhakar/Nandi/SR/Panduit or better) duly grooving concealed in the walls. All the required accessories such as Deep boxes, Jn Boxes, Normal bends, diversion bends etc are to be arranged by the contractor,as specified by supervisor in-charge.	RMT	2350	78.00	183300.00
20	20	Supply & fixing of PVC Flexible pipe of 25/40/50 mm dia sizes of Reputed make as specified by supervisor in-charge.	RMT	1200	60.00	72000.00
21	21	Supply & fixing of 4 modular PVC/MS box with top plate covers on the walls/Tables as specified by supervisor in-charge & Necessary fixing materials are to be arranged by the contractor. Make Pointer/Roma/Legrand	Numbers	600	127.00	76200.00
22	22	Supply and fixing of PVC channels (Ganapati/Shreeram/star/Legrand/Sudhakar/panduit make) with size of 1 mtr length, 100 mm x 50mm with top cover and laying the cables. Necessary fixing materials are to be arranged by the contractor. (3 anchor bolts per metre to be fixed),as specified by supervisor in-charge.	Metre	150	595.00	89250.00
23	23	Supply & installation of CT Boxes with wago terminals (50 pair) of WAGO make. consist of part nos:280-870 (100nos),280-374(1nos),249-116(2nos),210- 229/PW11-0000 (1nos) and Metallic box for outdoor use with locking facilit y-1 nos.	Numbers	31	4165.00	129115.00
24	24	Supply & fixing of 8 way RJ-45 Jack suitable for computer on the modular PVC box including termination of cable pairs as specified by supervisor in-charge.Make Pointer/Roma/Legrand	Numbers	40	400.00	16000.00
25	25	Supply and fixing of Steel mesh to arrest the rat entries in to cable path as as specified by supervisor in-charge. (All required materials are to be arranged by the contractor)	Kg	100	250.00	25000.00
26	26	Supply and painting (including spray painting) the S&T gadgets with approved quality of paint manufactured by Asian/British/ Johnson & Nicholson different colours of enamel paints at various	Sqm	1850	78.00	144300.00

		locations as per the instructions of JE/SE/SSE in-charge of site.				
27	27	Manufacturing and supply of engineering grade retro reflective sticker of EC socket post of numbering of size 8cm X 50cm . Numbering will be given by Railways. Characters and number in arial black font. (Make: 3M or Avery Denison Retro reflective sticker)	Numbers	400	90.00	36000.00
28	28	Laying of G.I Pipes /DWC size of 25/50/65/100mm dia.or defferent sizes over cement/girder bridges /FOB /COP / walls, platform etc., as per the instructions of site incharge. Necessary suitable MS clamps are to be provided with all fixture and supports etc. All materials except GI pipes are to be arranged by the contractor.	Metre	800	91.00	72800.00
29	29	Digging out the existing thermoshrinkabe joint made for Derivation from the ground & cable, opening the joint, cleaning the cable ends, making the cable loose by removing the loops made in the ground.as per instructions of site incharge.	Numbers	290	600.00	174000.00
30	30	Cutting of trench across/on the platform/ floor with cutting Machine for laying the cable through PVC/GI pipe and filling the trench after cable laying and restoring the surface with similar materials to match with existing floor/PF as per instructions of site in charge.	Metre	1650	250.00	412500.00
31	31	Excavation of pit, casting concrete foundation and erection of apparatus case full size as per Drg. No. SK/CN/1/89/A/1 and fixing of 1 No. of 'E' type lock for the front door, fixing of one hylam sheet of 1000mm X 940mm X 10mm thickness and painting the apparatus case inside and outside with one coat of redoxide and two coats of aluminum paints (foundation bolts and nuts,' E' type lock, cement, river sand, stone jelly of size 20/20mm, hylam sheet of size 1000mm X 940mm X 10mm, paints, varnish, fixing bolts and nuts and all other miscellaneous material required for the work shall be supplied by the contractor).	Numbers	16	9875.00	158000.00
32	32	Blowing of armoured Optical Fiber Cable into HDPE duct by using blowing machine and associated works .Before blowing the duct integrity shall be tested .If any defect shall be found during duct integrity test, the entire length of HDPE duct from one chamber to another chamber chamber is to be replaced by Contractor at free of cost .	Kilometre	8	8256.00	66048.00
33	33	Excavation for Jointing pit and provision of joint chamber of 1.2 mtr dia of 60 cm height with 2 piece cover on top and bottom cover with holes for drainage (reinforcement drawing will be as per technical specification) .This includes water cooling of chamber , backfilling and ramming of trench after laying and consolidated of soil as well as disposal of soil as per plan and as per instructions of site incharge	Numbers	36	5240.00	188640.00
34	34	Transporting and supply and installation of Concrete (RCC) route indicators at intervals of 50 meters as per instructions of site incharge. They should be of standard RCC with letters "SCR SIG/SCR 6Q cable" as per type of cable laid to drawing . In case of telecom quad cable joint locations it shall be painted with red color.	Numbers	150	807.00	121050.00

35	35	Shifting of coach indication board including fixing frame without damage from one place to another COP/post and wiring as per instruction of site incharge. Required fixing materials are to be arranged by the contractor.	Numbers	30	520.00	15600.00
36	36	Shifting of Train indication board including fixing frame without damage from one place to another COP/post and wiring as per instruction of site incharge. Required fixing materials are to be arranged by the contractor.	Numbers	30	520.00	15600.00
37	37	Releasing of EC post/Rail post, work involves grooving of posts/poles by digging/cutting to the required depth and Handing over to engineer incharge	Numbers	30	500.00	15000.00
38	38	Strengthening of foundation for EC socket post by carrying out the earth work around the foundation, ramming of the earth, carrying out masonry work from bottom of earth work using country stones and cement masonry to prevent the earth from slipping down the bank. (All required materials are to be arranged by the contractor	Numbers	120	500.00	60000.00
39	39	Strengthening of foundation for EC socket post by carrying out the earth work around the foundation, ramming of the earth, carrying out masonry work from bottom of earth work using country stones and cement masonry to prevent the earth from slipping down the bank and painting & provision of refractive stickering of reputed make and as per the instructions of Engineer Incharge	Numbers	120	824.00	98880.00
40	40	Fixing of horn/column/box type speakers/ CCTV cameras with 6mm thickness flat angles to the Electrical posts/COP available on the platform with suitable coupling arrangements as per the instructions of site incharge	Lumpsum	60	150.00	9000.00
41	41	Demolishing R.C.C. work including cutting and stacking of steel bars and disposal of unserviceable material within 50m lead.as per instructions of site incharge	cum	20	2534.70	50694.00
42	42	Concreting of portion of the pipe at the bend near the culvert including cement asper the drawing SK.15/11. (Item unit is "Perbend ") as per instructions of site incharge	Numbers	200	1216.00	243200.00
43	43	Excavation of coil pit of size 1.5mtrX1.5mtr and depth of 1.00 mtrs for providingof 6 quad cable in coil at joint locations as per the instructions of site incharge	Numbers	140	124.00	17360.00
44	44	Splicing of Optical Fiber Cable with splicing loss should be less than 0.5db with neat dressing in FDMS/Joint Enclosure as per instructions of site supervisor (Fully loaded) All required materials are to be arranged by the contractor	Pair	1000	180.00	180000.00
45	45	Laying of 5 pair screened sheathed cable/ power cable/LS cable/UTP/ PVC cables through PVC Pipe as per instruction of site incharge. Pipes will be supplied by Railways.	Metre	66800	5.00	334000.00
46	46	Rectification and re splicing of existing mid section OFC joint. This includes supply and provision of joint repair kit & sleeves. All materials including splicing machine used for fiber splicing and, OTDR for	Numbers	24	6210.78	149058.72

		testing and other testing materials if any shall be arranged by the contractor				
47	47	Installation of 2 wire DTMF Selective Calling Telephone with power supply.as per instruction of site incharge	Numbers	20	611.00	12220.00
48	48	Excavation of earth and providing earth pipe of size 50mm dia, 3.5mm thick 2.0 m. long G.I. pipe with masonry tub constructed around at earth locations as specified by site Engineer. Alternate layers of 30kgs-chrcoal and 30kgs-common salt to be filled in the pit while closing. The pipe perforation etc., to be carried out as per drawing No. G/SG/09/2002. The earth pipe to be connected to the function with cable as supplied by Railways. The reading should be painted on the masonry tub. The masonry tub of 300X300X300mm should be covered with cement slab. (All materials to be supplied by the contractor except cable.)	Numbers	60	5100.00	306000.00
49	49	Design and preparation of cable route plan, cable core integrated cable route plan in defferent colours (Quad & OFC cables etc), other Telecom wiring diagrams for IPIS, PA system, CCTV etc. in in A-4 size/-4Nos (A2 size) sizes as required with 1 set of prints on tracing film with approval, and a soft copy on compact disc (CD/pendrive) to be supplied. and as per instruction of site incharge .(Unit : per sheet (A2 size))	Sheet	360	600.00	216000.00
50	50	Arranging of Print (in Color) of A4/A3/A2 size cable path diagrams and other indoor cable path drawings as specified by Engineer Incharge.and tracing film drawings will be arranged by railways	Sheet	6000	11.80	70800.00
51	51	Binding of A4/A3/A2 size cable path color printed diagram (sectionwise) into booklets as specified by Engineer incharge drawings will be arranged by railways. Unit : Booklet)	Numbers	48	35.40	1699.20
52	52	Laying of U/G cable (PIJF Copper cable/OFC Cable) on the outdoor structures such as over bridges/walls through GI Pipes wherever required. For fixing the above, required fixing materials such as clamps, screws, etc are to be arranged by the contractor as specified by supervisor in-charge. GI pipes will be supplied by Railways.	RMT	1000	100.00	100000.00
53	53	Supply and fixing of PVC channels (Ganapati/Shreeram/star/similar make) with size of 1 mtr length, 150 mm x 50mm with top cover and laying the cables.Necessary fixing materials are to be arranged by the contractor. (3 anchor bolts per metre to be fixed),as specified by supervisor in-charge.	RMT	50	815.00	40750.00
54	54	Supply and fixing of PVC channels (Ganapati/Shreeram/star/similar make) with size of 1 mtr length, 50 mm x 50mm with top cover and laying the cables. Necessary fixing materials are to be arranged by the contractor. (3 anchor bolts per metre to be fixed),as specified by supervisor in-charge.	RMT	400	405.00	162000.00
55	55	Splicing of Optical Fiber Cable (2x 12 fibers) in / FDMS/LIU/ Termination boxes and testing and splicing loss should be less than 0.5db.(All required materials are to be arranged by the contractor)	Numbers	50	4000.00	200000.00

56	56	Supply& installation of 100 pair cable termination MS BOX/Block for terminating 100 pair cable containing 10 Nos. of 10 pair LSA Plus termination strips with rubber gronet with universal keys,as specified by supervisor in-charge.Make: Krone /bestnet/ Airtel.	Numbers	20	2725.00	54500.00
57	57	Supply & installation of 200 pair cable termination BOX for terminating 100 pair cable containing 20 Nos. of 10 pair LSA Plus termination strips with rubber gronet with universal keys ,as specified by supervisor in-charge.Make: Krone /bestnet/ Airtel.	Numbers	20	6500.00	130000.00
58	58	Supply & installation of 400 pair cable termination BOX for terminating 100 pair cable containing 40 Nos. of 10 pair LSA Plus termination strips with rubber gronet with universal keys as specified by supervisor in-charge.Make: Krone /bestnet/ Airtel.	Numbers	10	9800.00	98000.00
59	59	Supply and jointing of thermo shrink jointing kit for 10/20/50/100 pair UG PIJF cables as per RDSO Specn RDSO/SPN/TC/57/2015 Rev.1 or latest and as per instructions of site incharge.	Numbers	10	2900.00	29000.00
60	60	Supply of 4 way type RJ-11 Dual Jack suitable for telephones fixed on the modular PVC top covers including fixing and termination of cable pairs. Make Pointer/Roma/legrand or similar	Numbers	650	180.00	117000.00
61	61	Supply of 6 way type RJ-11 Jack suitable for telephones on the modular PVC top cover including fixing and termination of cable pairs.Make Pointer/Roma/legrand or similar	Numbers	250	130.00	32500.00
62	62	Supply of 4 way type RJ-11 Jack suitable for telephones fixed on the modular PVC top covers including fixing and termination of cable pairs. Make Pointer/Roma/legrand or similar	Numbers	2200	93.00	204600.00
63	63	Supply & fixing of Dummies on PVC modular boxes 2/4/6/8 modulars.Necessary fixing materials are to be arranged by the contractor. Make Pointer/Roma/legrand or similar	Numbers	500	25.00	12500.00
64	64	Supply & fixing of 2 modular PVC/MS box with top plate covers on the walls. Necessary fixing materials are to be arranged by the contractor. Make Pointer/Roma/legrand or similar	Numbers	500	110.00	55000.00
65	65	Supply of 8/6 Modular PVC/MS box with top cover on the walls.Necessary fixing materials are to be arranged by the contractor. Make Pointer/Roma/legrand or similar	Numbers	400	137.00	54800.00
66	66	Supply and fixing of TV Socket on 2 modular PVC Box suitable fixing materials are to be arranged by contractor make: Pointer/Roma/ Legrand.	Numbers	150	68.00	10200.00
67	67	Supply & fixing of Co axial cable coupler of reputed make for extension of TV connections in the Chambers and as specified by supervisor in-charge.	Numbers	50	75.00	3750.00
68	68	Supply and installation of rack mounted 12F FMS including 12Nos 0.2db SC /PC adaptors and 12 numbers 900 micron 1.5 meter SCPC pigtails. Make Raychem/Bestnet/Dlink or better	Numbers	20	3500.00	70000.00
69	69	Supply & fixing of 15U rack. With locking front glass door and fully vented rear steel door, compatible with standard 19" accessories with cooling fan,	Numbers	1	18000.00	18000.00

		power manager with 5 sockets,1 equipment tray, earth connecting kit etc this includes wiring as per the instruction of site incharge. Make: HCL/ Vero president /Dlink/Bestnet				
70	70	Supply & installation of Rack 19" wall mount 6U/500 MM D with front glass door and lock and key (Make: HCL/Vero president/bestnet) including the following: (1) 1U cable manager - 1 No. (2) 1U Cantilever tray - 1 No.(3) AC distribution box 5X5 amps - 1 No.(4) Front panel H/W PKT of 10 Sets - 1No. this includes wiring as per the instruction of site incharge. Make: HCL/ Vero president/bestnet.	Numbers	10	6400.00	64000.00
71	71	Supply & fixing of 4U rack. With locking front glass door and fully vented rear steel door, compatible with standard 19" accessories, top cover with gland plate and fan, with rack cage nuts and screws, fan assembly kit, rack mount shelves, Power manager.this includes wiring as per the instruction of site incharge. Make: HCL/Vero president/ Bestnet.	Numbers	4	5400.00	21600.00
72	72	Supply & Installation of 6A/16A Dual pole MCB of Make:Legrand of similar and as specified by supervisor in-charge.	Numbers	10	600.00	6000.00
73	73	Supply and installation of EXIR 4MP outdoor Bullet IP camera POE ONVIF Network CCTV. Security WDR IR 30M English Version 6mm Lens 1 M Histosino cable with power adaptor Make: Hikvision model DS-2CD2T43G0-i5 or similar and as specified by supervisor in-charge .	Numbers	12	9676.00	116112.00
74	74	Supply & Installation 6TB HDD suitable to surveillance system NVR/DVR Make:Segate/Wdisk or better	Numbers	6	12100.00	72600.00
75	75	supply and installation of SC to SC 0 db Connector suitable for FMS 24F telescopic 19 inches Rack. Make MX/ Bestnet.	Numbers	100	225.00	22500.00
76	76	Shifting/replacing of set top boxes of DTH/ Cable connections in Officers chambers, Rest house with neat dressing of cables, as per the directions of site incharge.	Job	10	850.00	8500.00
77	77	Supply and fixing of glass stand with stainless steel brackets. (Size 12InchX10 inch X 6mm thicknes) and Necessary fixing materials such as screws wooden pegs etc All required materials are to be arranged by the contractor.	Numbers	10	2200.00	22000.00
78	78	Fiber splicing and termination along with neat dressing in LIU and as per instructions of site in charge (Required materials including joint enclouser are to be arranged by the contractor	Pair	1500	88.50	132750.00
79	79	Supply ,Installation, integration and commissioning of ONVIF 16 channel NVR 4 SATA supports up tp 6 TB of similar make Dahua/Hikvision/CP plus to model No.4416-4KS2 or better	Numbers	2	27000.00	54000.00
80	80	Supply and installation of SIP Video Phones Model T49G Make : Yealink or CoreIP or Tadiran Telecom or Grand Stream.	Numbers	6	49560.00	297360.00
81	81	Supply and Fixing of IO Boxes make D-Link/HCL/Bestnet	Numbers	300	192.34	57702.00
82	82	Supply and installation of SIP Basic Phones Model: T31P or better Make : Yealink/CoreIP/Tadiran	Numbers	10	4767.20	47672.00

		Telecom/Grand Stream.				
83	83	Supply and installation of SIP Excutive Phones Model: SIP-T43U or or better Make : Yealink/CoreIP/Tadiran Telecom/Grand Stream.	Numbers	4	10856.00	43424.00
					Total	9067990.12